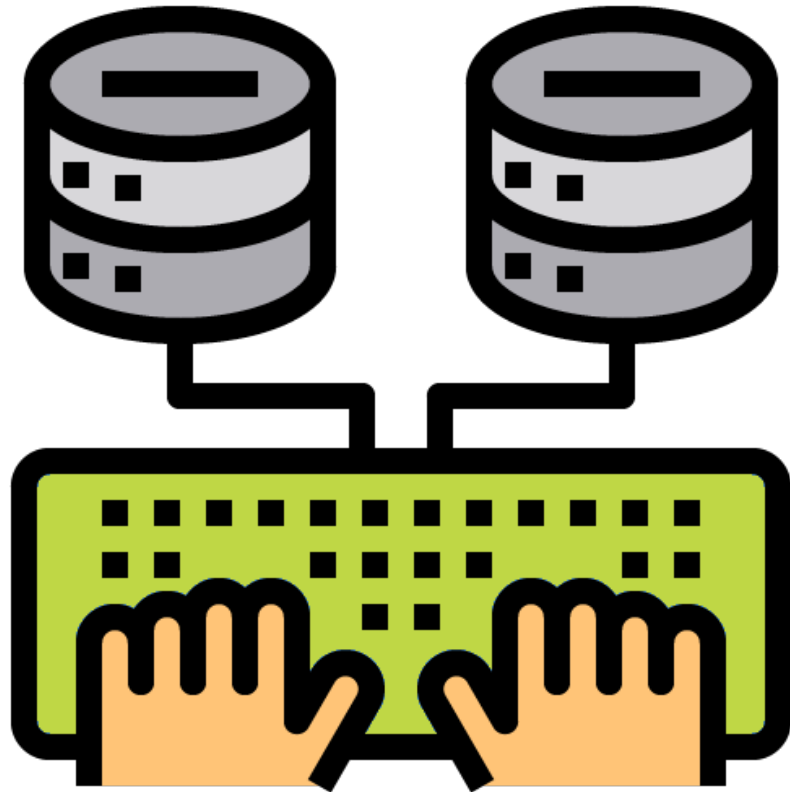




Big Data Analytics

Big Data Platforms

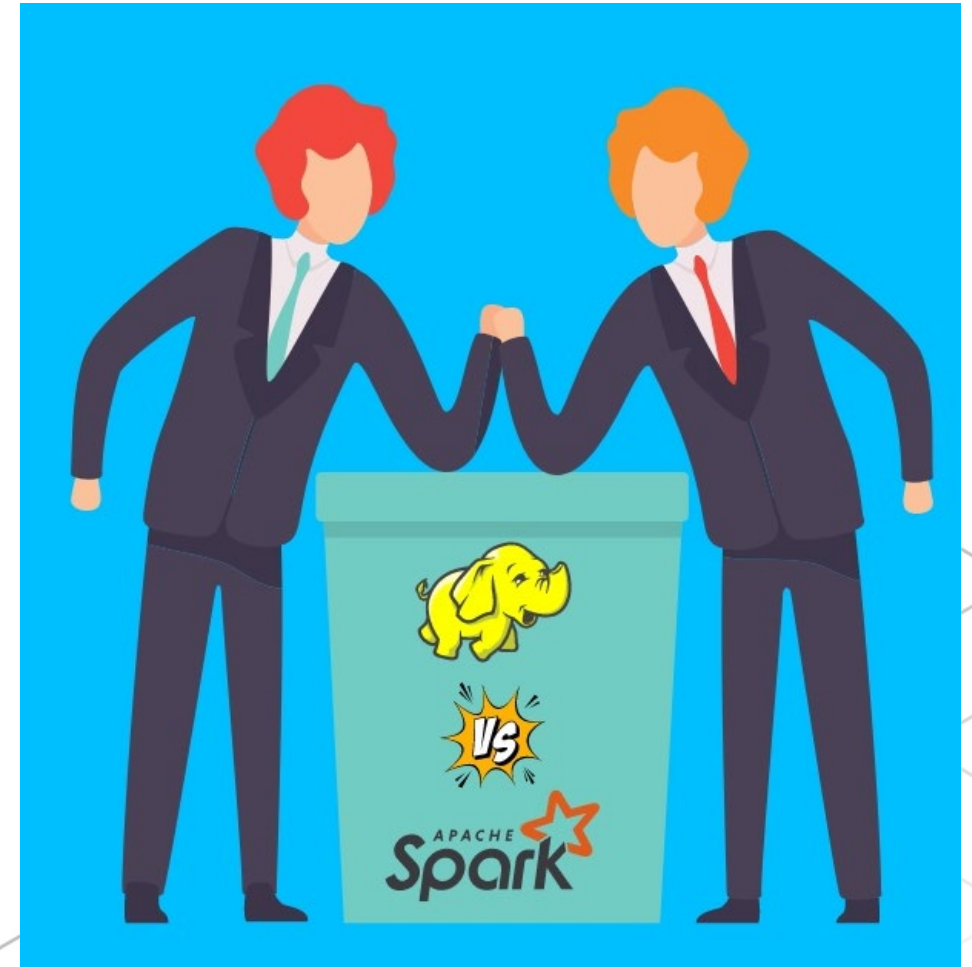
Lecturer:

Márcia Lourenço Baptista
m.baptista@novaims.unl.pt

Learning Objectives

By the end of this lecture, students should be able to:

- Understand big data platforms (Hadoop, Spark, cloud).
- Explain and understood Hadoop as the original big data platform



Congratulations!!

- You were hired by **Nova Cidade Lab** to Prepare the Infrastructure for Traffic Management Using Big Data & Distributed Computing



Scenario

- Imagine Lisbon facing severe traffic congestion. Let us develop an **AI-powered Smart Traffic System** to optimize traffic flow.
- However, this requires processing **real-time big data** from:

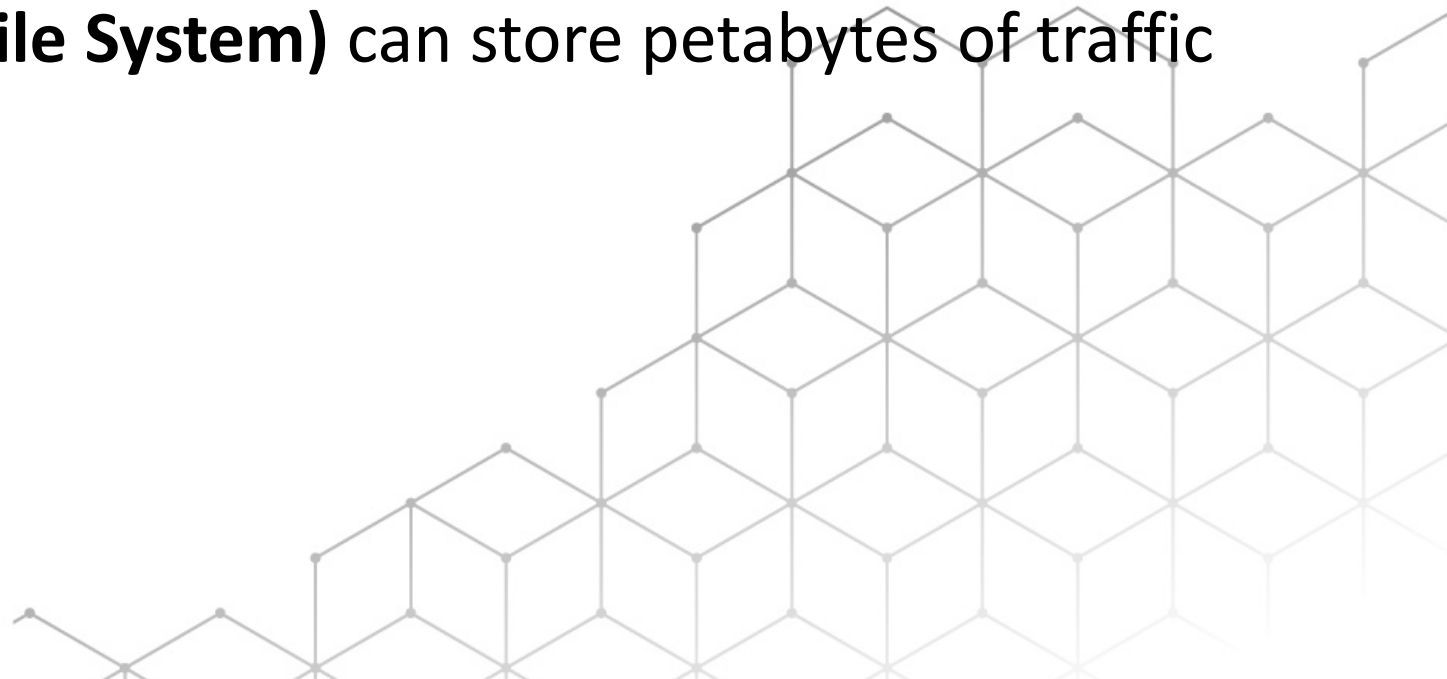


- A single server **cannot** handle this efficiently. **Big Data platforms like Hadoop, Spark** come into play.

Challenge 1: Historical Records

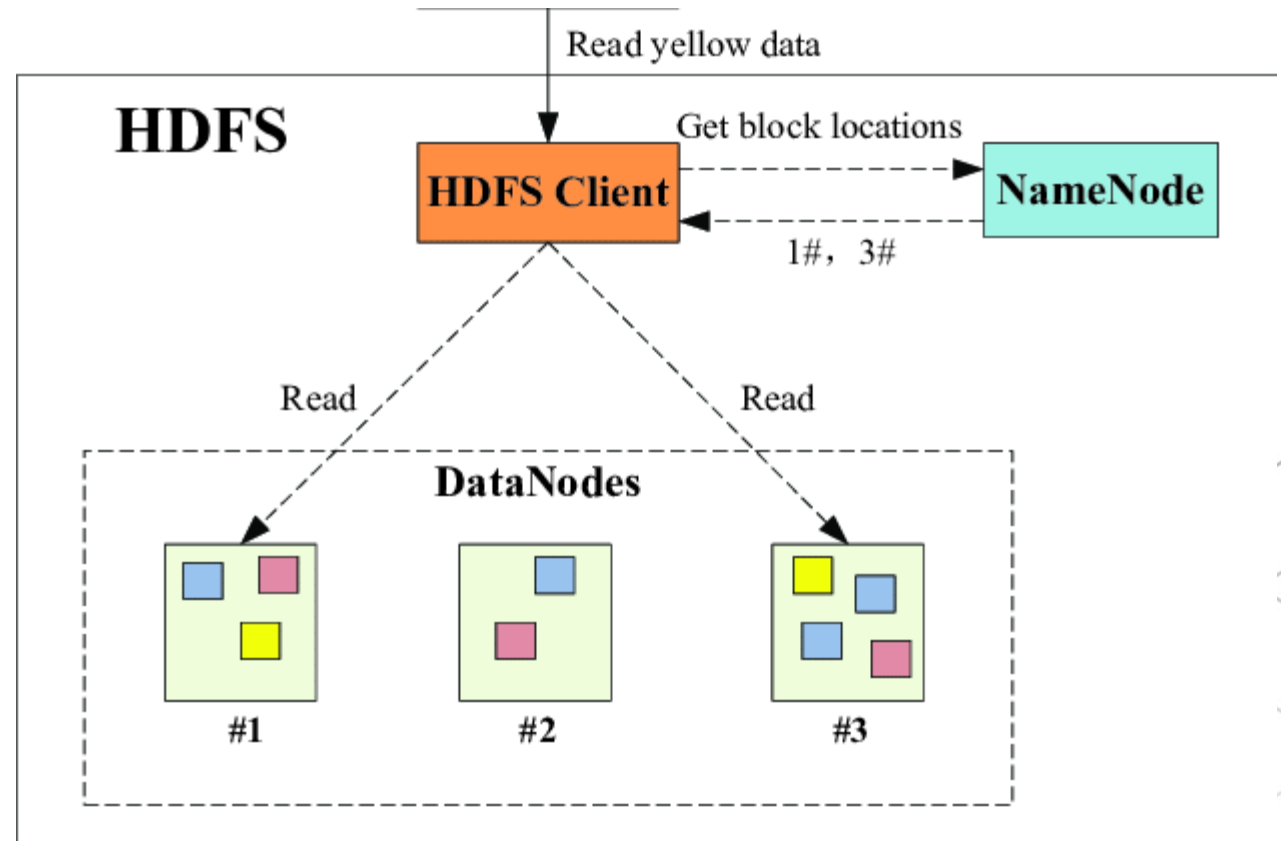
Challenge: The city has years of historical traffic data stored in unstructured logs, images, and GPS records. This data must be **stored** somewhere.

- **HDFS (Hadoop Distributed File System)** can store petabytes of traffic data across multiple servers.



What is a Distributed File System?

- A **Distributed File System (DFS)** is a system that allows to access and manage files stored across a network of interconnected storage nodes as if they were on a single local file system.



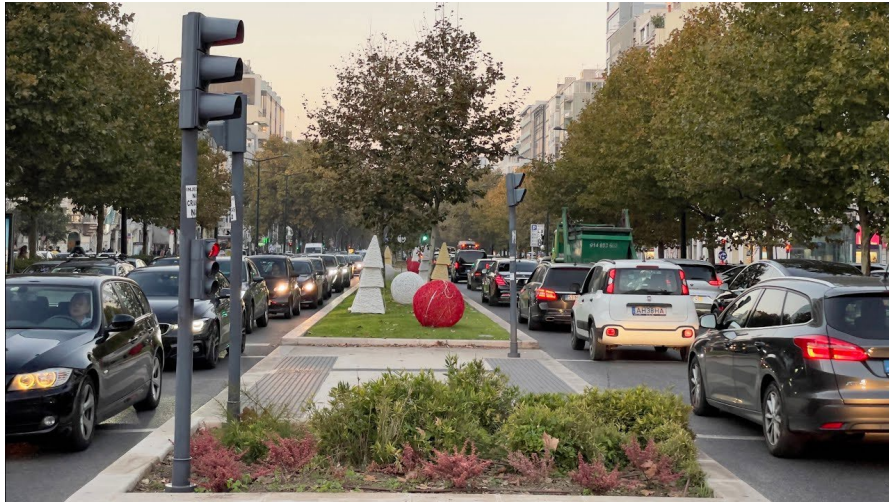
Challenge 2: Real-Time Traffic Monitoring

- The system must process **live traffic** and predict congestion.
- **Spark** is used for **data streaming** and **AI predictions**.
 - **Spark Streaming** processes **data from millions of vehicles every second**.
 - **Machine Learning (MLlib)** predicts traffic congestion.
 - **GraphX** helps analyze city road networks to find the best paths.



Challenge 3: Adapt to Big Data

- **Challenge:** handle **millions of data points** during peak hours.
- **Manage Spark clusters dynamically** to ensure smooth performance.
 - **Autoscaling** – If traffic data spikes, **automatically add computing power**.
 - **Fault tolerance** – If a server fails, **redistribute workloads instantly**.



Rush Hour



Scale Down

Challenge 4: Using Cloud Services

The city doesn't want to buy and maintain expensive servers.

- **Cloud-Based Big Data Platforms**

Example:

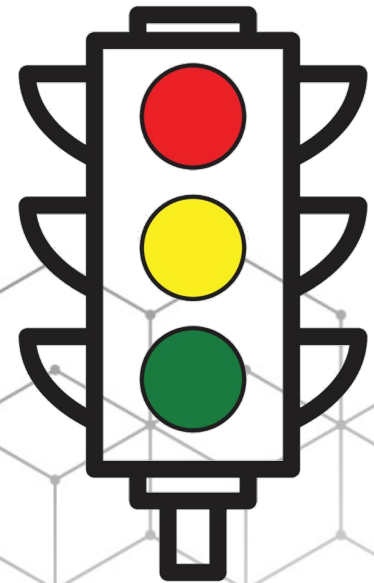
- The city **rents cloud servers only when needed**, reducing costs.
- Cloud services integrate AI for **predicting congestion**.



Potential Benefits

Instead of **static traffic lights**, AI intersections could:

-  **Use real-time traffic demand** to adjust who stops.
-  **Communicate with connected vehicles** to flow.
-  **Prioritize emergency vehicles** by clearing their path.



Why are Big Data Platforms Important?

- **Scalability:** Handle massive datasets from diverse sources. (Volume)
- **Speed:** Enable real-time data analysis. (Velocity)
- **Integration:** Seamless collaboration with tools and systems. (Variety)
- **Costs:** Reduce infrastructure costs by distributed computing.



Agenda for Today...

- Big Data Platforms (Hadoop, Spark)
 - Big Data Platform
 - Components of Big Data Platforms
 - On-Premise vs Cloud Big Data Platforms
 - Serverless vs Traditional Cloud Big Data Platforms
 - Hadoop vs Spark
- Hadoop
- Next Gen Hadoop



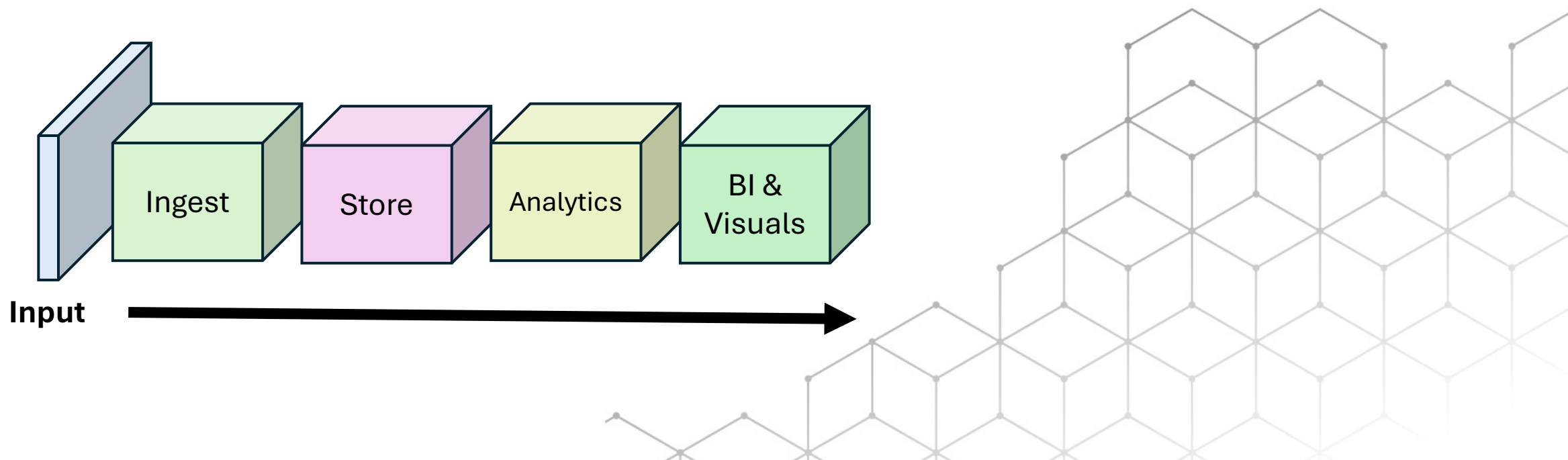
Questions

- What is a **Big Data Platform**?
- What are the **key components** of a Big Data Platform?
- What are the differences between **on-premise and cloud-based Big Data Platforms**?
- Give me examples of **Big Data Platforms**?
- What is serverless cloud computing?
- Give me examples of (open-source) **Big Data Platforms**?

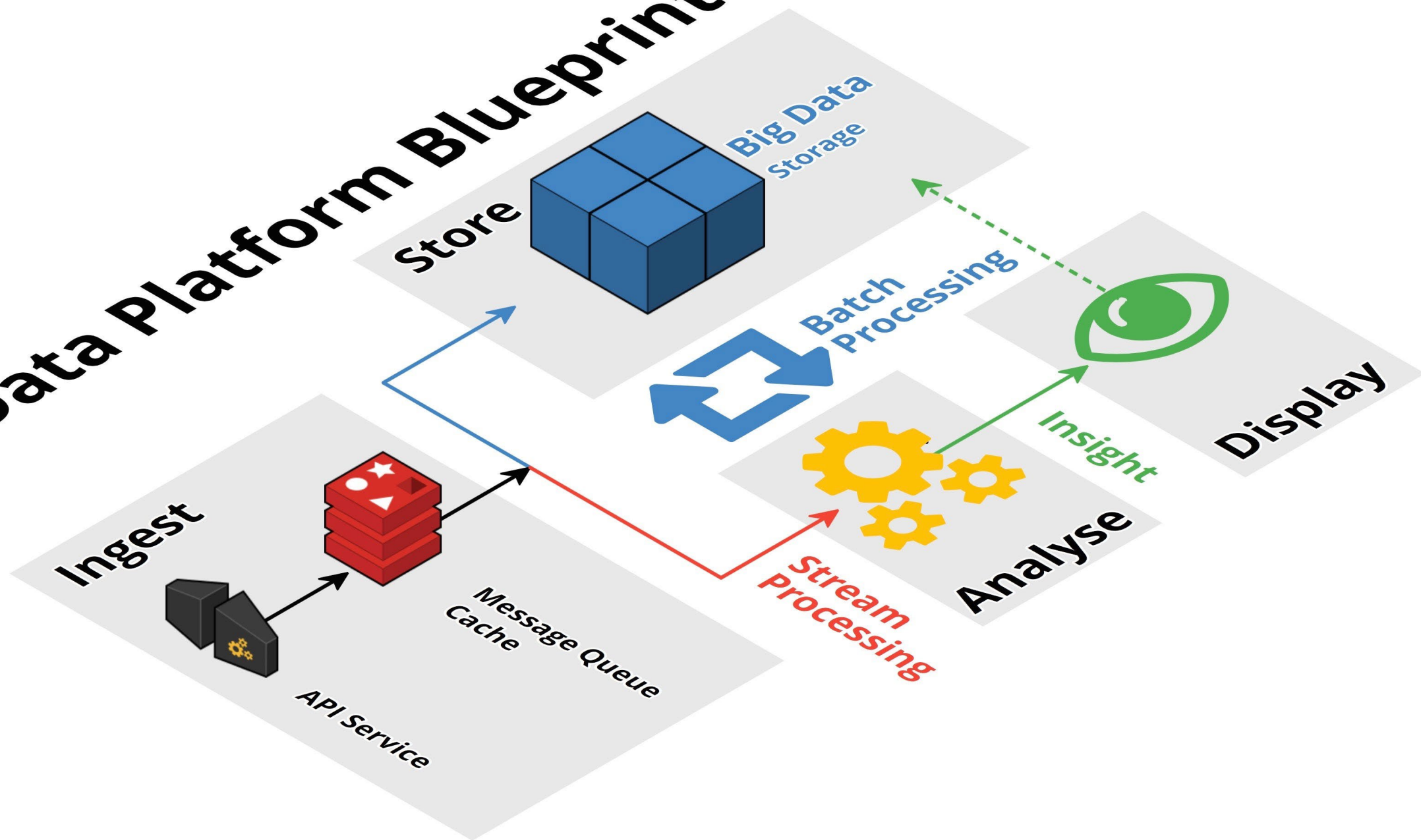


Big Data Platform

- A **Big Data Platform** is an integrated system designed to ingest, store, analyze, and display big data.
- The platform consists of several key components that work together to handle data.



Big Data Platform Blueprint



Main Components of Big Data Platform

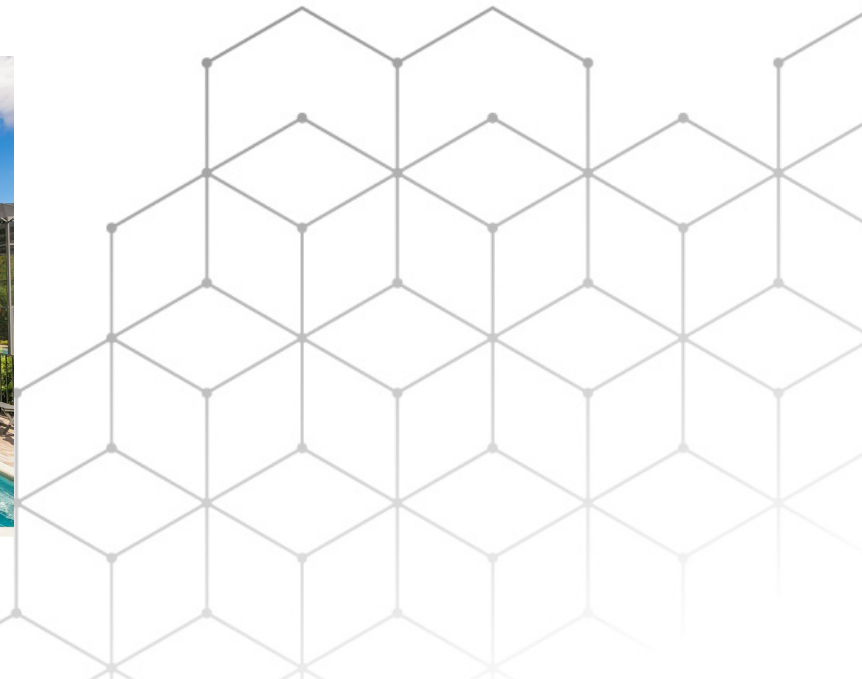
- **Data Sources (Input Layer):** Where data comes from
- **Data Ingestion:** Brings raw data into the system
- **Data Storage (HDFS, NoSQL, Cloud Storage)**
- **Data Analytics:** Extracts insights & predictions from data
- **Data Visualization & Business Intelligence**
- Can you think of more components?



On-Premise vs Cloud Big Data Platforms

Big Data Platforms can be deployed in two main ways:

- **On-Premise** – Managed **internally** using own servers.
- **Cloud-Based** – Hosted on **third-party cloud providers** (AWS, Azure, or Google Cloud).



On-Premise = Owning a House

- You **buy and maintain** the house (servers).
- You **pay upfront costs** (hardware, setup).
- You **handle repairs & upgrades** (maintenance).
- You have **full control** but **limited flexibility**.



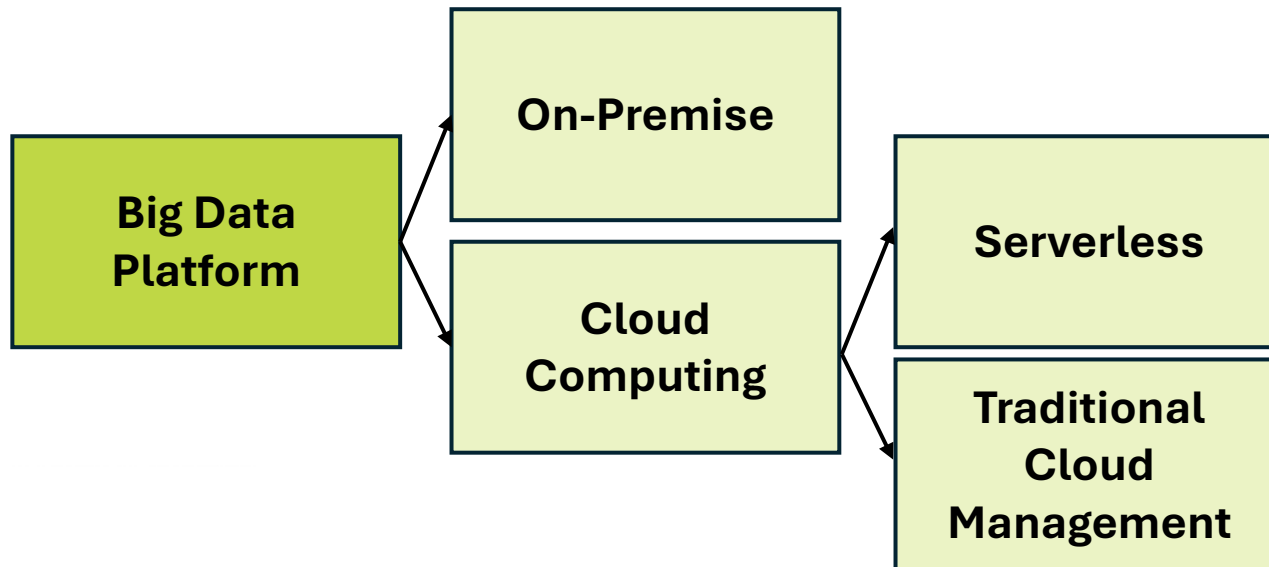
Cloud-based = Renting an Apartment

- You **pay monthly rent** (pay-as-you-go model).
- The **landlord (cloud provider)** handles **maintenance**.
- You can **move to a bigger apartment anytime** (scalability).
- Less control over customization but **more convenience**.



Types of Cloud-Based Big Data Platforms

- **Serverless computing** is a cloud computing model where we **run applications without managing servers**.
- A cloud provider (AWS, Google Cloud, Azure) handles everything
- Users **only pay for the actual execution time**.



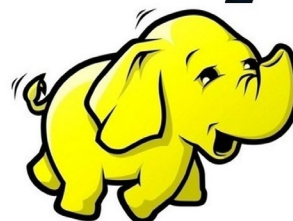


Popular Big Data Platforms

- Apache Hadoop:

- Open-source platform for distributed storage and batch processing.
- Core components:
 - HDFS (storage),
 - MapReduce (processing),
 - YARN (resource management).

hadoop



- Apache Spark: Unified engine for large-scale data processing.

- Fast in-memory computations and supports MLlib, GraphX, and Spark SQL.



time

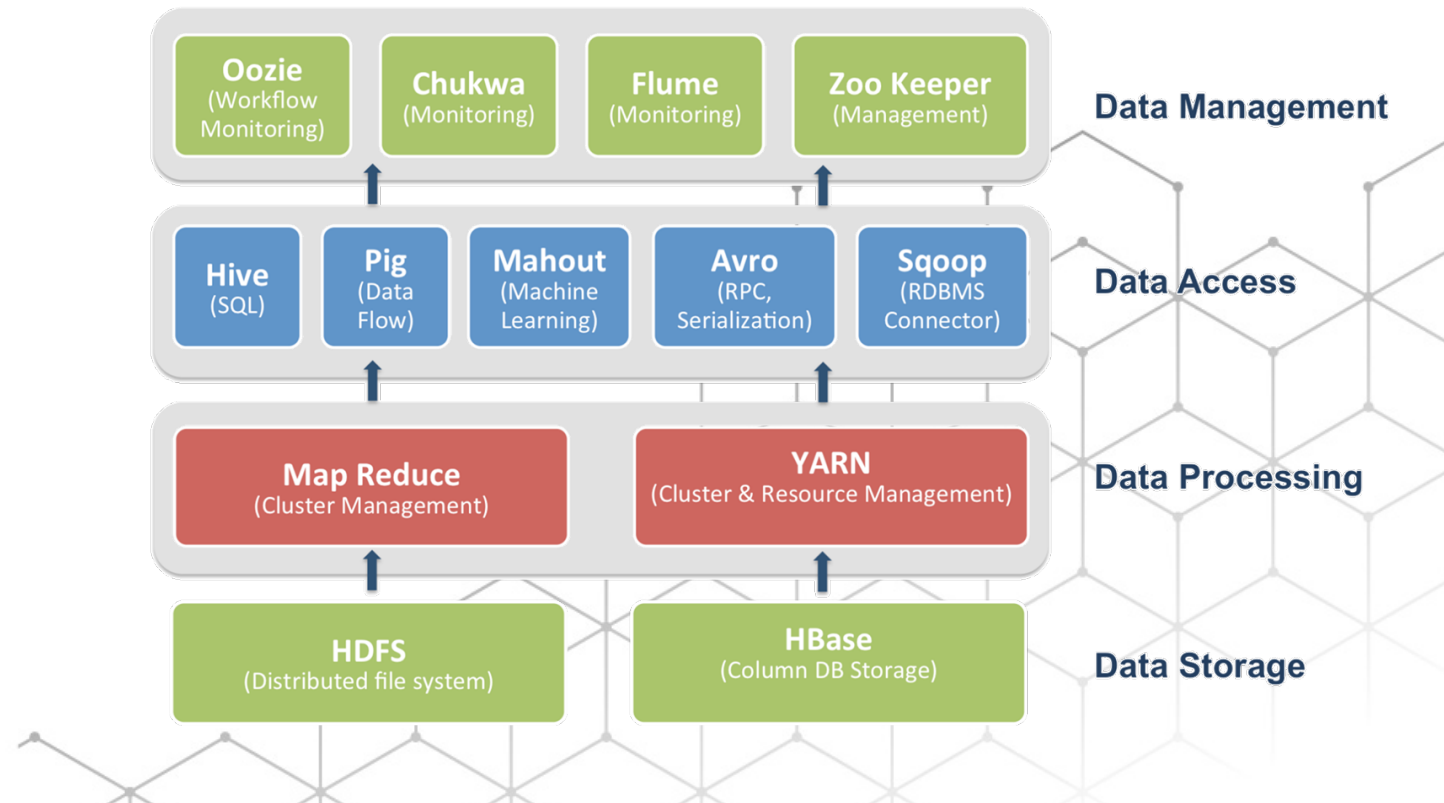


Hadoop: Original Big Data Platform

The Hadoop platform provides on-premise and cloud deployment options for large-scale data processing.

Apache Hadoop Ecosystem

- **HDFS** – Distributed storage.
- **MapReduce** – Batch data processing.
- **YARN** – Resource management.
- **Apache Hive** – SQL-like querying.

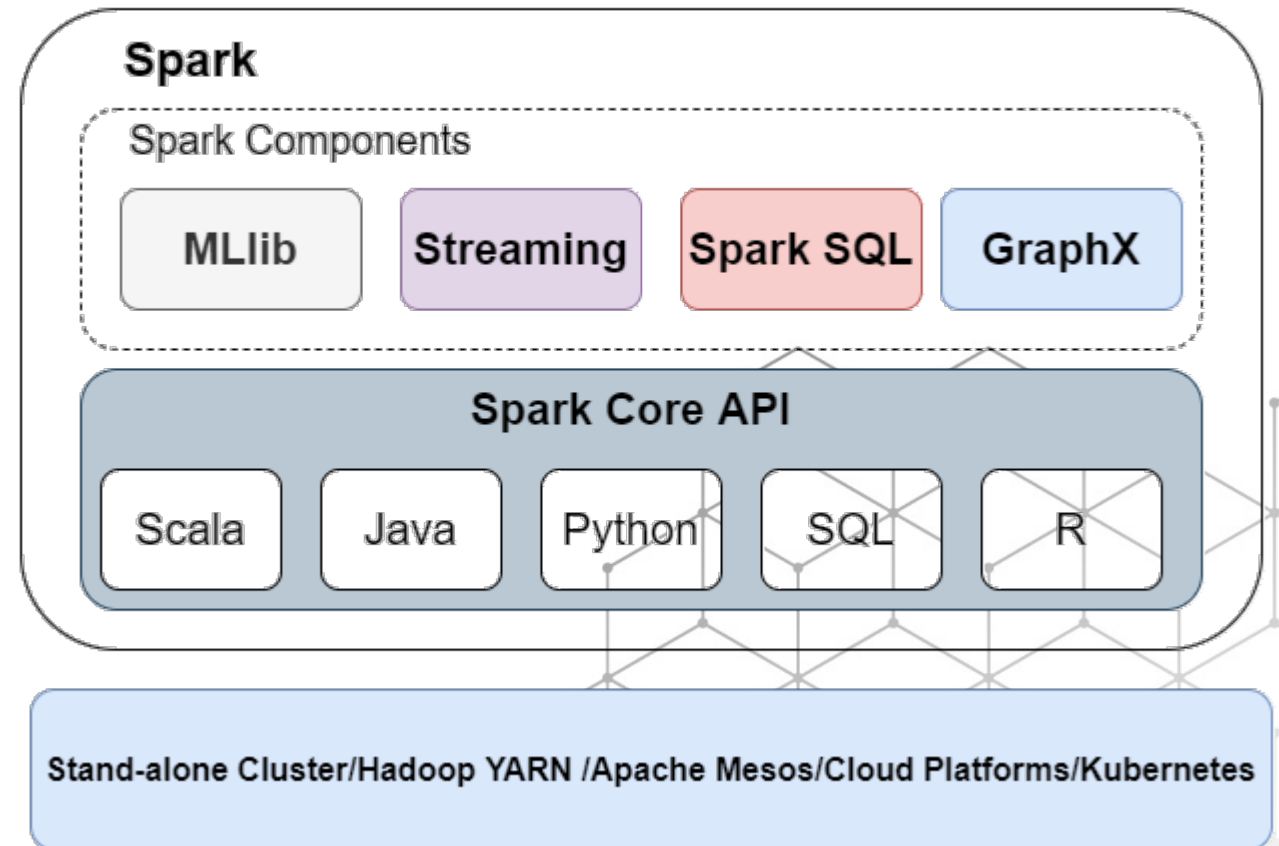


Open-Source Apache Spark

Apache Spark

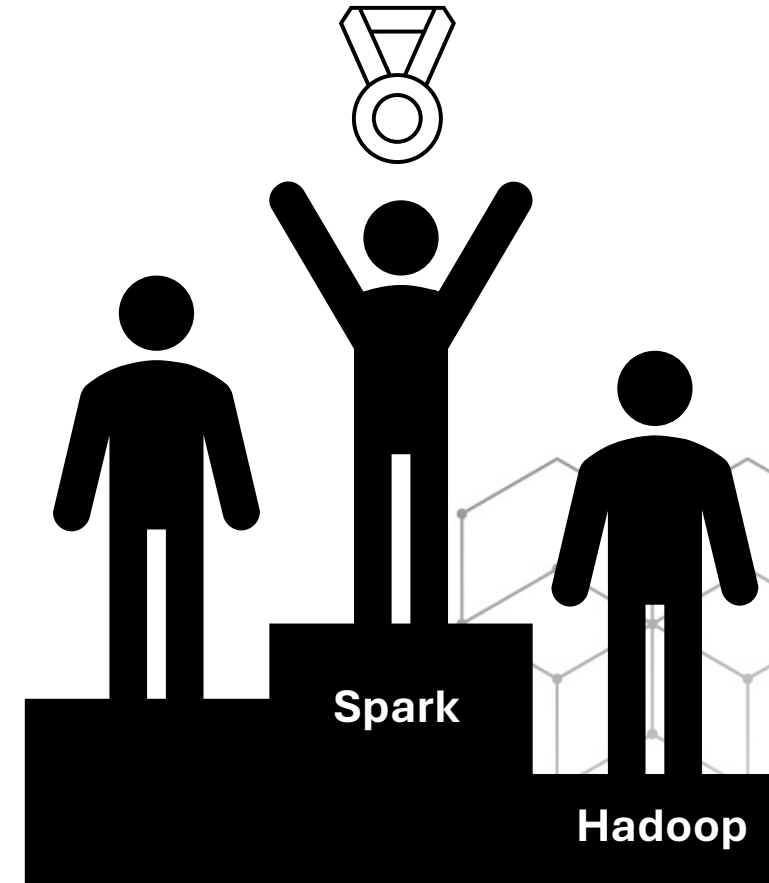
- **In-memory processing** (faster than Hadoop).
- Supports **SQL, ML, Graph processing, and Streaming**.
- **Databricks** is a cloud-based version of Spark.

Spark Ecosystem



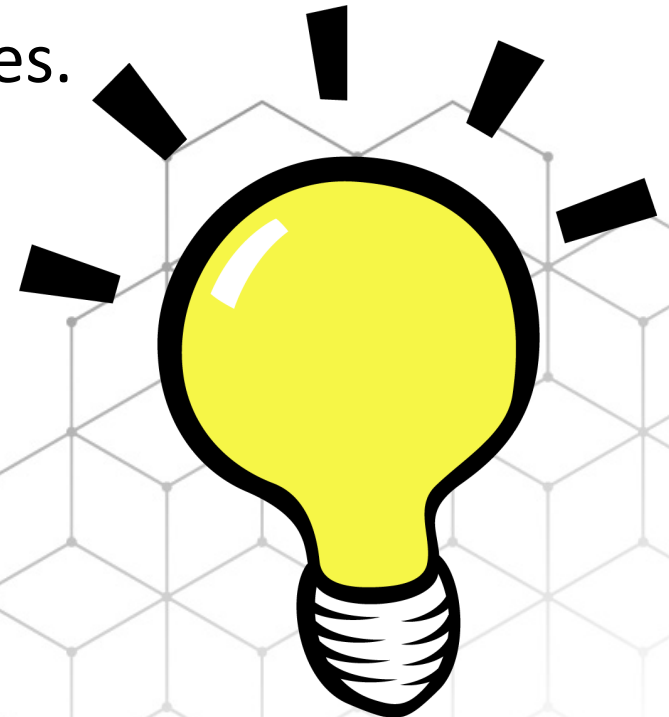
Is Hadoop a Realistic Alternative?

- While Hadoop was once the **dominant technology**
- Hadoop is **still used**, but its role in Big Data **has changed significantly**
- **Legacy systems & batch processing.**
- **Declining** in real-time streaming and AI workloads.



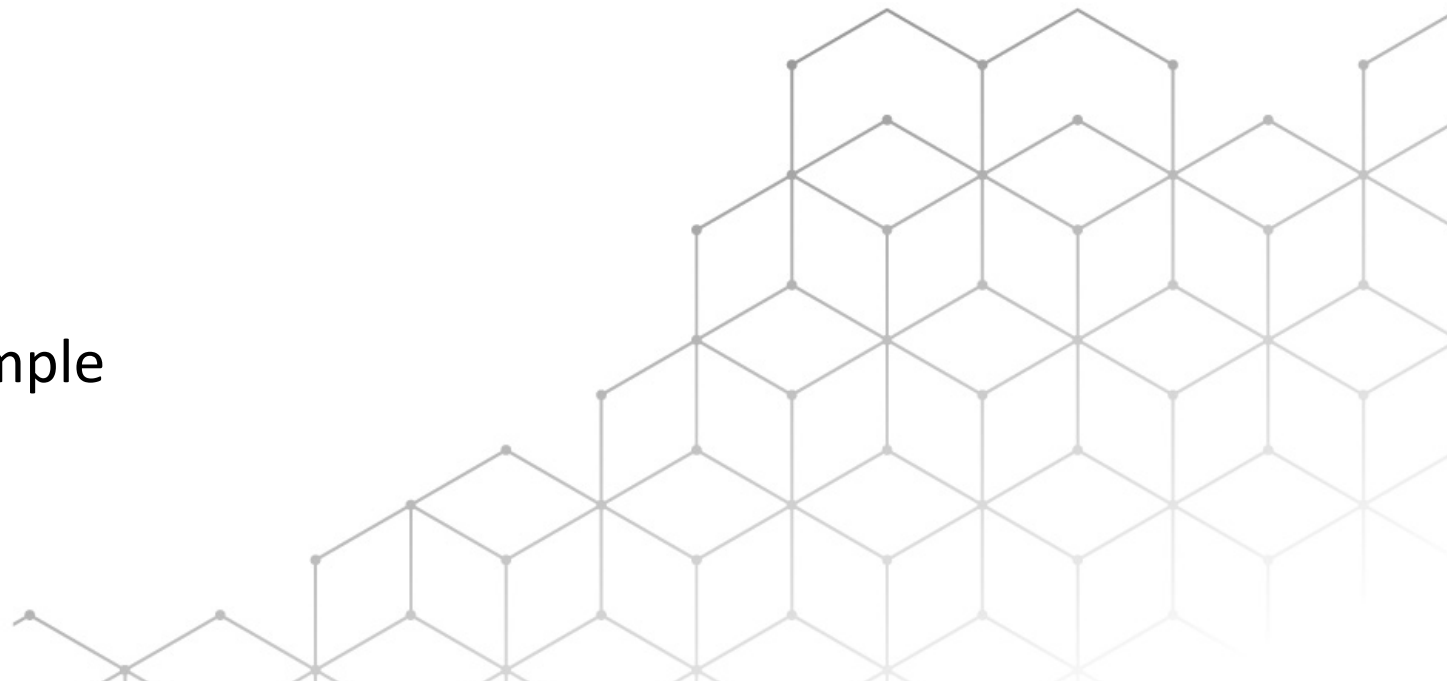
Brief Note: What is Databricks?

- Cloud-native, serverless Apache Spark with built-in optimizations.
- Great for AI, ML, and data engineering.
- Eliminates Hadoop cluster management headaches.
- Integrates with Delta Lake for structured data processing.



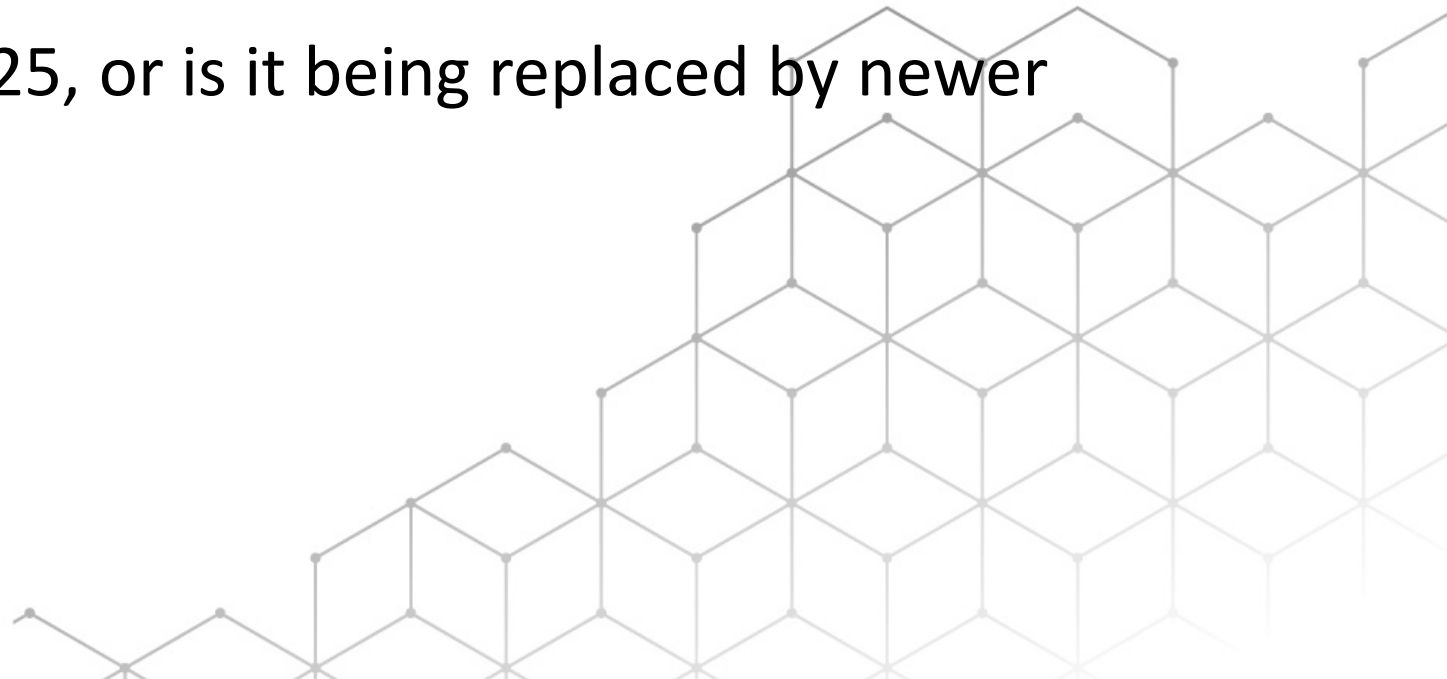
Agenda for Today...

- Big Data Platforms (Hadoop, Spark)
- Hadoop Platform
 - What is Hadoop?
 - History of Hadoop
 - Basic Components of Hadoop
 - Hadoop Ecosystem
 - Hadoop (Main) Distributions
 - Goals in Hadoop 1.0 and 2.0
 - MapReduce Word Count Example
- Next Gen Hadoop



Questions

- Why was Hadoop created, and what problem does it solve?
- How does Hadoop allow companies to store and process petabytes of data efficiently?
- Why is Hadoop considered a "distributed" framework?
- Is Hadoop still relevant in 2025, or is it being replaced by newer technologies?

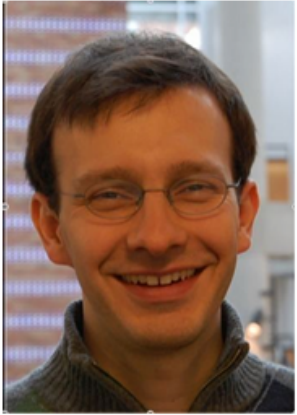


What is Hadoop?

- Hadoop is an **open-source** framework (software library), written in Java, for big data **processing** and **storage**.



Apache Nutch



**Mike
Cafarella**



**Doug
Cutting**

worked together between 2002/2004 on



goal: build a search engine system that can index 1 billion pages?



Google Seminal Papers in 2003 and 2004



released two seminal papers in 2003 and 2004

The Google File System

Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung
Google*

ABSTRACT

We have designed and implemented the Google File System, a scalable distributed file system for large distributed data-intensive applications. It provides fault tolerance while running on inexpensive commodity hardware, and it delivers high aggregate performance to a large number of clients.

While sharing many of the same goals as previous distributed file systems, our design has been driven by observations of our application workloads and technological envi-

1. INTRODUCTION

We have designed and implemented the Google File System (GFS) to meet the rapidly growing demands of Google's data processing needs. GFS shares many of the same goals as previous distributed file systems such as performance, scalability, reliability, and availability. However, its design has been driven by key observations of our application workloads and technological environment, both current and anticipated, that reflect a marked departure from some earlier

MapReduce: Simplified Data Processing on Large Clusters

Jeffrey Dean and Sanjay Ghemawat

jeff@google.com, sanjay@google.com

Google, Inc.

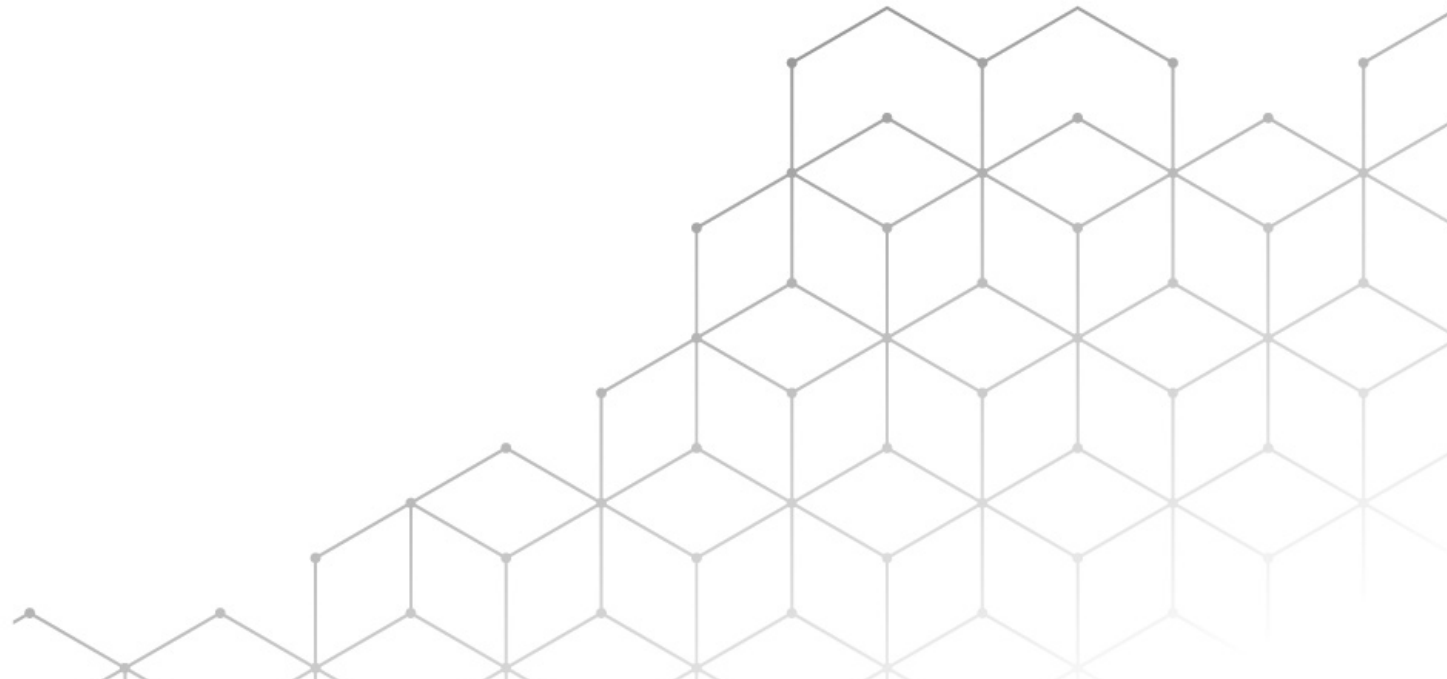
Abstract

MapReduce is a programming model and an associated implementation for processing and generating large data sets. Users specify a *map* function that processes a key/value pair to generate a set of intermediate key/value pairs, and a *reduce* function that merges all intermediate values associated with the same intermediate key. Many real world tasks are expressible in this model as shown

given day, etc. Most such computations are conceptually straightforward. However, the input data is usually large and the computations have to be distributed across hundreds or thousands of machines in order to finish in a reasonable amount of time. The issues of how to parallelize the computation, distribute the data, and handle failures conspire to obscure the original simple computation with large amounts of complex code to deal with

Contributions

- 2003 paper (GFS) offers a solution for **storing** very large files
- 2004 paper (MapReduce) offers a solution of **processing** those large datasets.



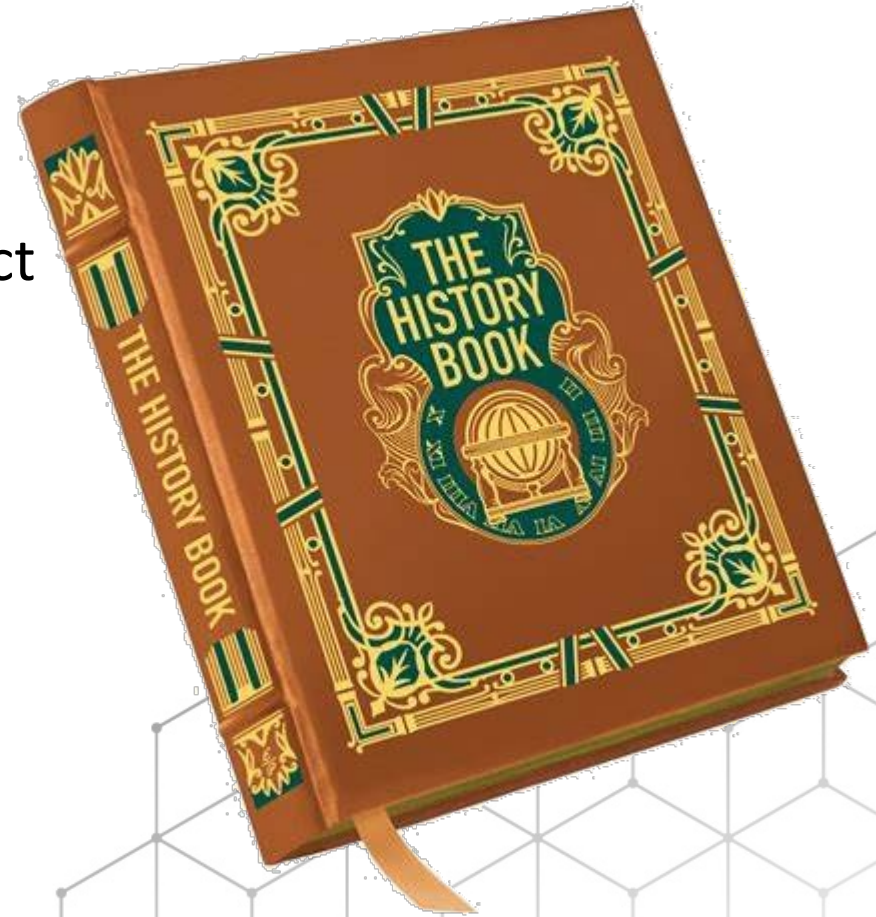
Doug's Hadoop Project

- in 2006 Doug Joins Yahoo and forms a new project called Hadoop



Timeline of Hadoop

- 2007 Hadoop successfully tested a 1000 node cluster
- 2008 Yahoo released Hadoop as an open-source project to Apache Software Foundation (ASF).
- 2009 Hadoop successfully sorted a PetaByte of data in less than 17 hours.
- 2011 Apache Hadoop V1.0 was released
- 2013 Apache Hadoop V2.06 was released
- 2017 Apache Hadoop V3.0 was released



Open Source



- **Yahoo** contributed the code for Hadoop to the broader software community under an **open-source license**
- ASF is a non-profit organization that oversees open-source software projects.
- Being open-source made Hadoop more popular, as companies like Facebook, LinkedIn, and Amazon adopted and improved it.

Basic Differentiators (1)

- Reliability: The Hadoop distributed filesystem offers replication of data, with a default **replication of 3x**.
- Flexibility: Variety of data including structured and **unstructured data** can be processed, stored, and analyzed.
- Cost effectiveness: Apache Hadoop is completely **open source**. Unlike traditional software, it can run on any hardware, and it does **not** require **high-end servers**;
- Scalability: Hadoop is a completely **distributed system**. With data growth, Hadoop clusters can add more nodes **dynamically** or even downsize them.

Basic Differentiators (2)

- Unlimited storage space: Storage in Hadoop can scale up to **petabytes** of data storage with HDFS.
- Unlimited computing power: Hadoop 3.x supports more than **10,000 nodes** (machines), Hadoop 2.x supports up to 10,000 node clusters.
- **Hadoop's complexity, maintenance costs, and slower performance** (compared to Spark) have led many companies to move away from traditional Hadoop.

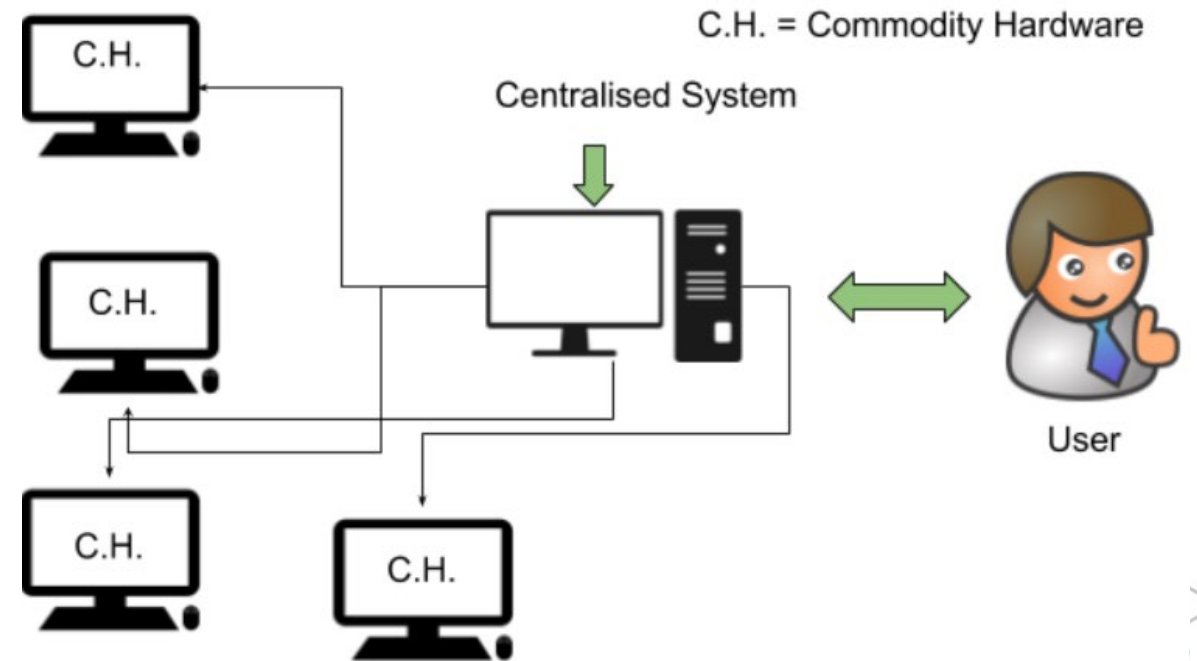
How Much is the Big Data in Hadoop?

- “Very large” in this context means files that are hundreds of megabytes, gigabytes, or terabytes in size.
- Petabytes of data.

Decimal		Binary		Rounded up percentage value for the difference between decimal and binary values
Bit (b)	0 or 1	Bit (b)	0 or 1	
Byte (B)	8 Bits	Byte (B)	8 Bits	
Kilobyte (kB)	1000 bytes	Kibibyte (KiB)	1024 bytes	2.40%
Megabyte (MB)	1000 Kilobytes	Mebibyte (MiB)	1024 Kibibyte	4.86%
Gigabyte (GB)	1000 Megabytes	Gibibyte (GiB)	1024 Mebibyte	7.37%
Terabyte (TB)	1000 Gigabytes	Tebibyte (TiB)	1024 Gibibyte	9.95%
Petabyte (PB)	1000 Terabytes	Pebibyte (PiB)	1024 Tebibyte	12.59%
Exabyte (EB)	1000 Petabytes	Exbibyte (EiB)	1024 Pebibyte	15.29%
Zettabyte (ZB)	1000 Exabyte	Zebibyte (ZiB)	1024 Exbibyte	18.06%
Yottabyte (YB)	1000 Zettabyte	Yobibyte (YiB)	1024 Zebibyte	20.89%
Ronnabytes (RB)	1000 Yottabyte	Not adopted yet (expected: <i>robibyte</i>)		23.79%
Quettabytes (QB)	1000 Ronnabytes	Not adopted yet (expected: <i>quebibyte</i>)		26.77%

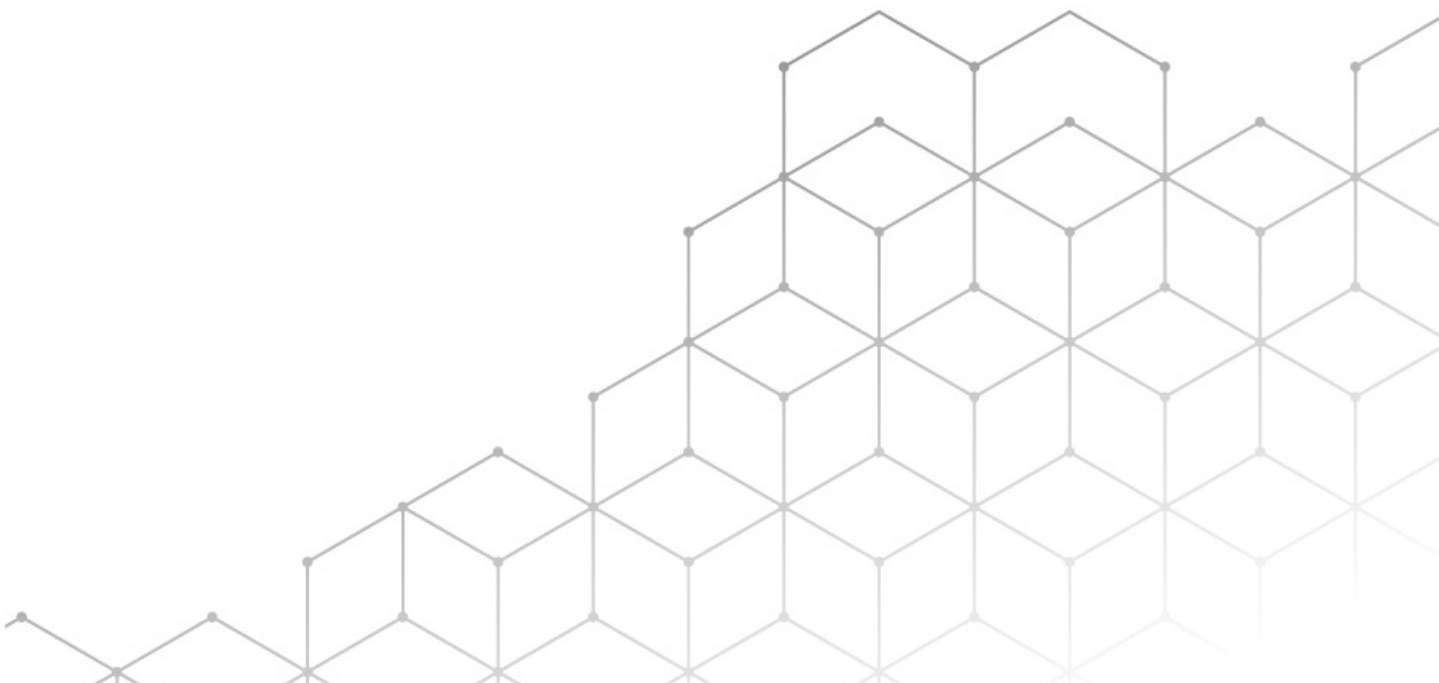
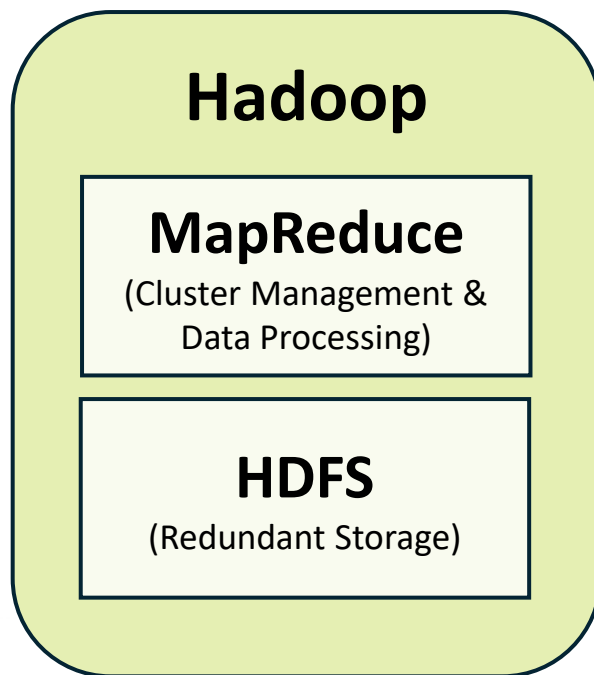
Commodity Hardware

- Hadoop doesn't require expensive, highly reliable hardware.
- It's designed to run on clusters of commodity hardware for which the chance of node failure across the cluster is **high**, at least for large clusters.
- HDFS is designed to **carry on** working without a noticeable interruption to the user in the face of such failure.



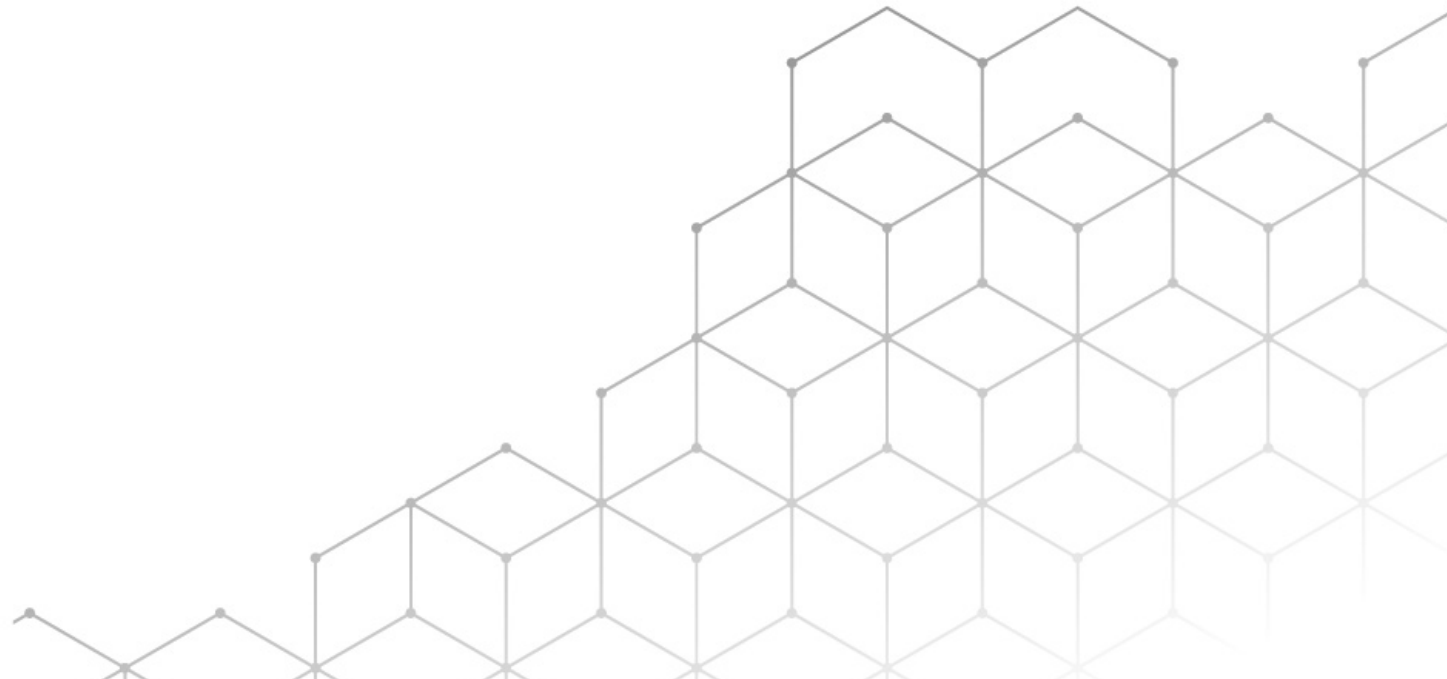
Original Components of Hadoop v1.0

- Distributed File System (HDFS): A distributed file system.
- MapReduce: Framework to process the data stored in HDFS



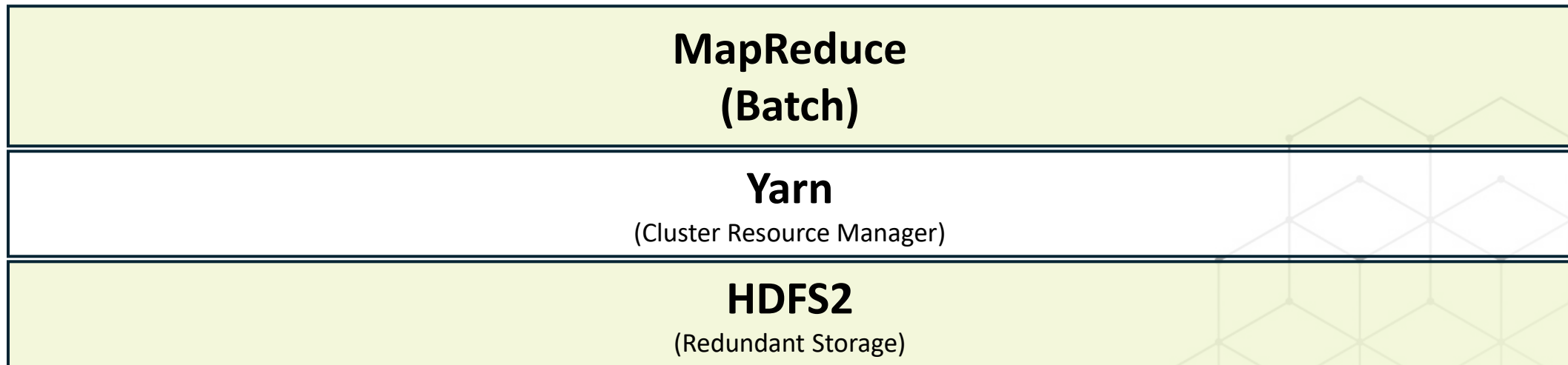
Components of Hadoop v2.0

- Since V2, Hadoop consists of three core components:
 - Hadoop Distributed File System (hdfs)
 - MapReduce
 - YARN



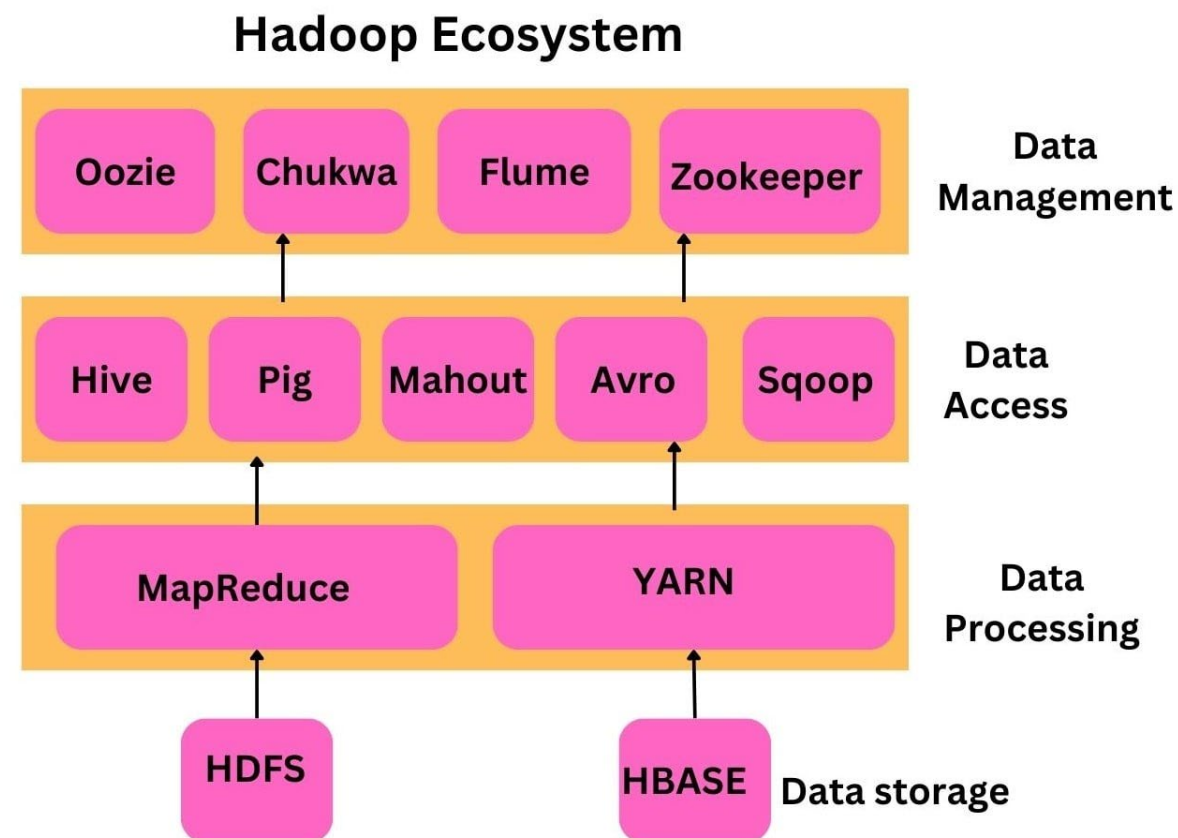
Yarn in Hadoop 2.0

- YARN functions as resource management layer within Hadoop.
- It supports various processing frameworks (not only MapReduce)



Hadoop Ecosystem

- Several modules have been added since Hadoop 1.0 (see right)
- Many are now independent projects on top of Hadoop.
- We are talking about more than 150 projects!

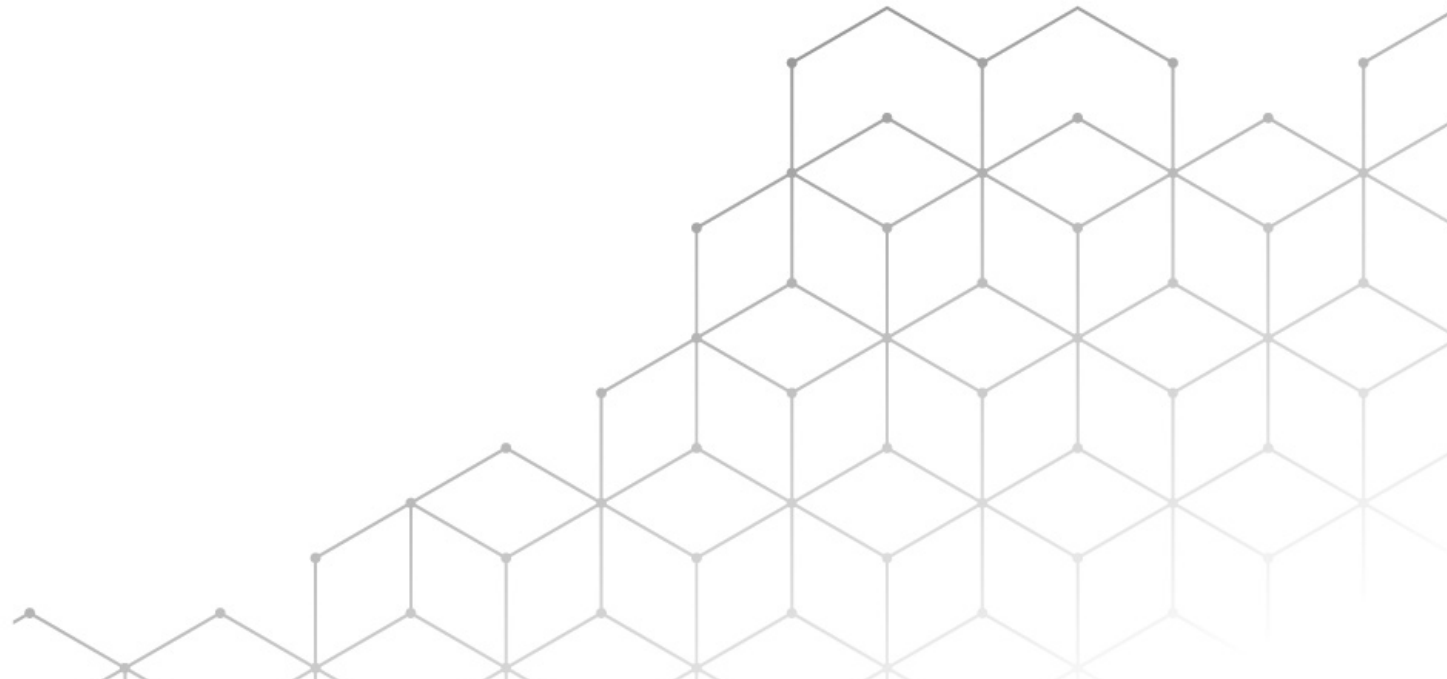


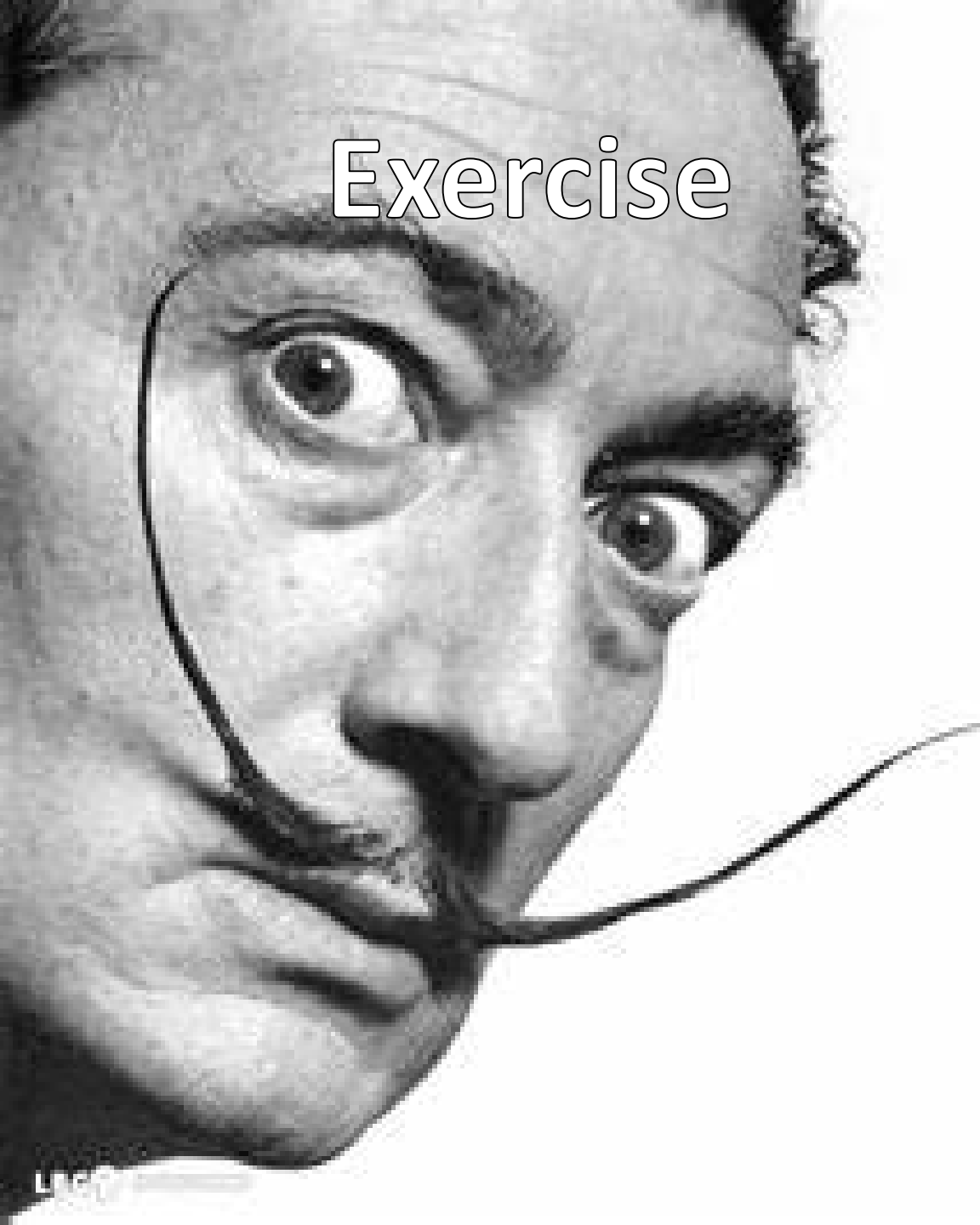
Hadoop (Main) Distributions

Hadoop Distribution	Advantages	Disadvantages
Cloudera	CDH has a user friendly interface with many features and useful tools like Cloudera Impala	CDH is comparatively slower than MapR Hadoop Distribution
MapR	It is one of the fastest hadoop distribution with multi node direct access.	MapR does not have a good interface console as Cloudera
Hortonworks	It is the only Hadoop Distribution that supports Windows platform.	The Ambari Management interface on HDP is just a basic one and does not have many rich features.

Agenda for Today...

- Big Data Platforms (Hadoop, Spark)
- Hadoop Platform
- Next Gen Hadoop





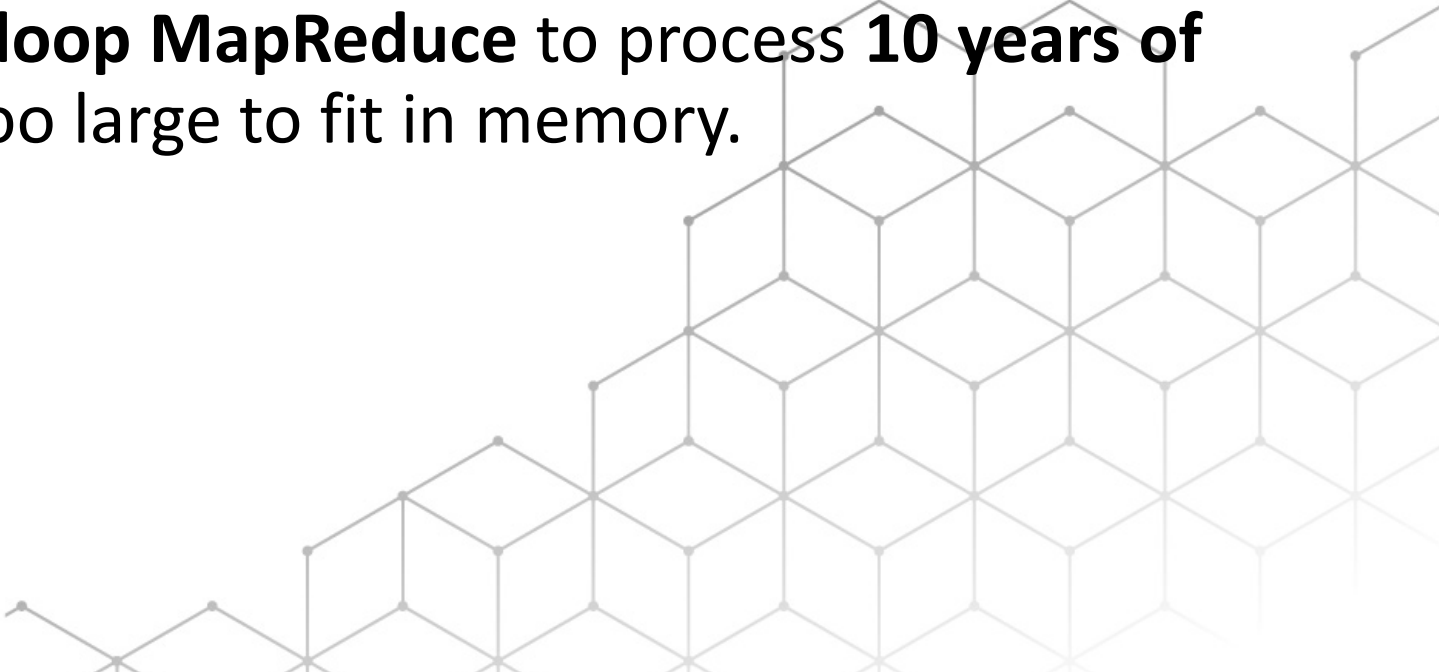
Exercise

- If you had to redesign Hadoop from scratch, what changes would you make?

Processing uses the Disk!

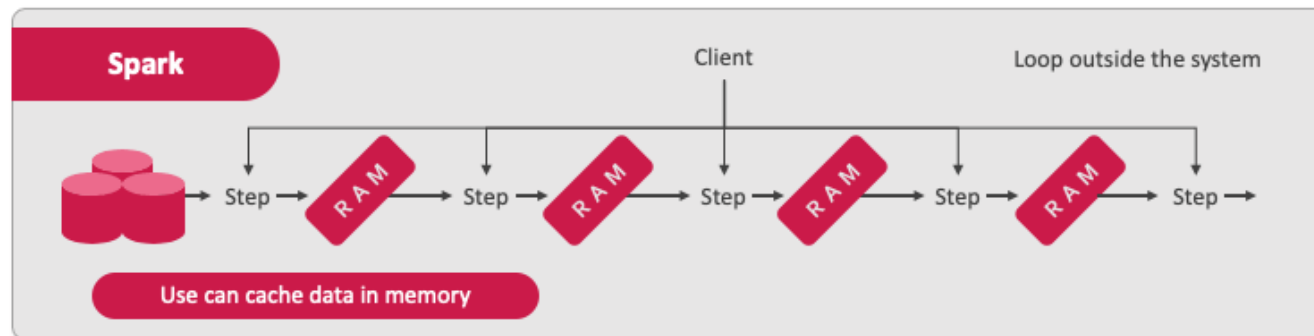
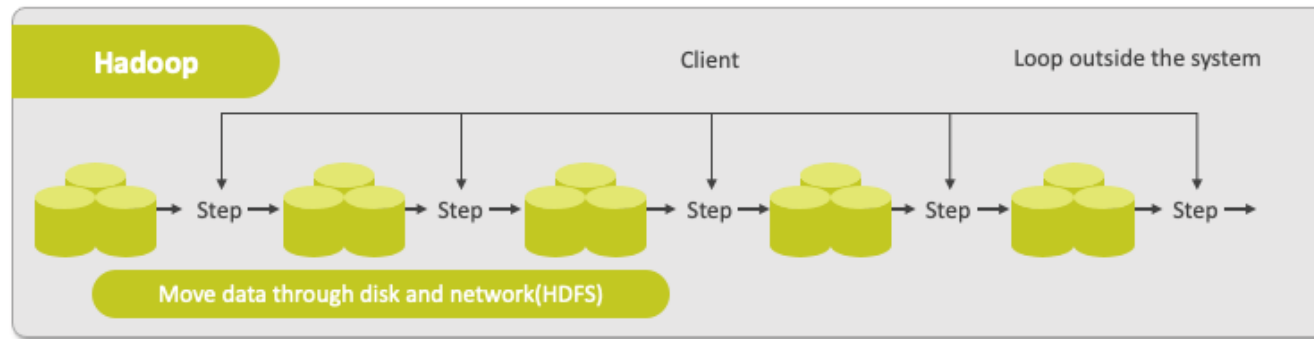
Hadoop MapReduce processes large files by reading chunks from **HDFS**, computing on them, and writing results back to disk before moving to the next step.

A financial institution uses **Hadoop MapReduce** to process **10 years of transaction history**, which is too large to fit in memory.



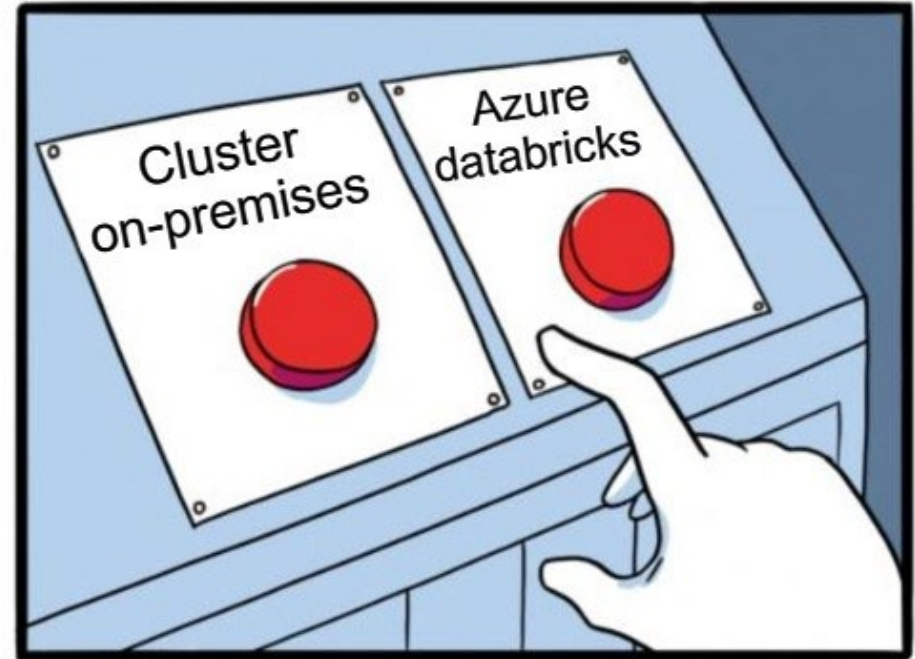
Processing Redesign

- Use in-memory processing instead of disk-based processing.
- Optimize for parallel execution like GPUs.



On-premises System!

Hadoop was designed for **on-premises distributed systems**, but modern applications are moving to **multi-cloud environments**.



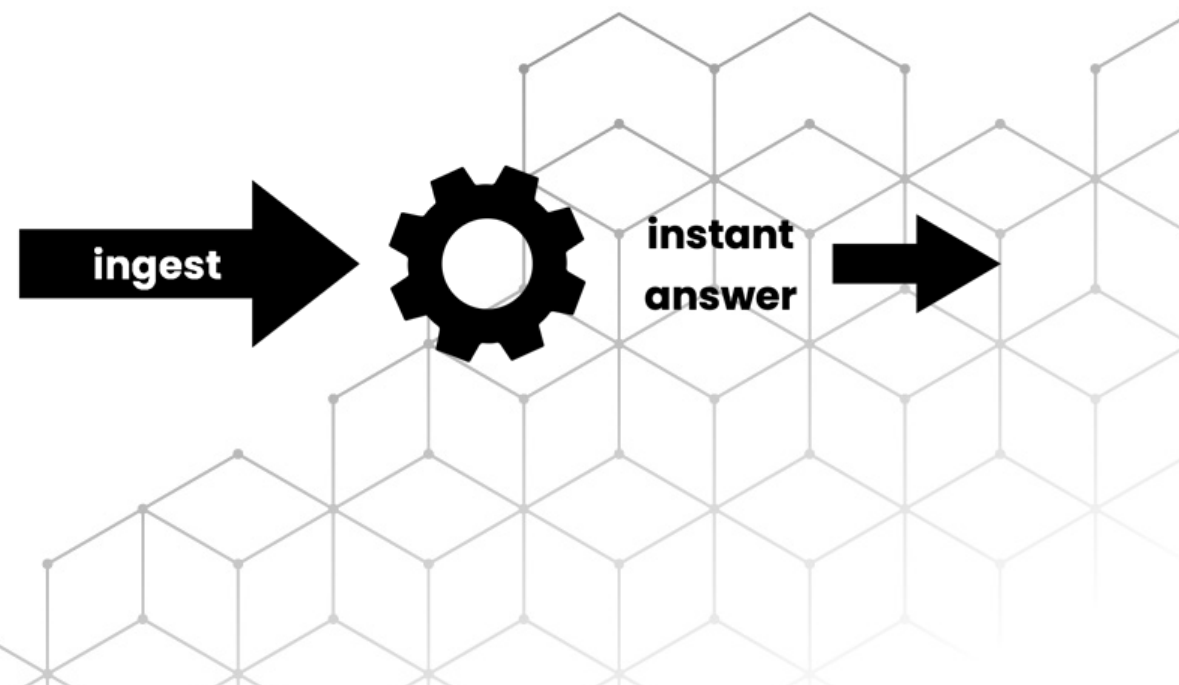
Issues

Issue: Hadoop is batch-oriented and **not ideal** for real-time streaming.

batch processing



stream processing



Manual tuning

- Hadoop cluster management **requires manual tuning.**

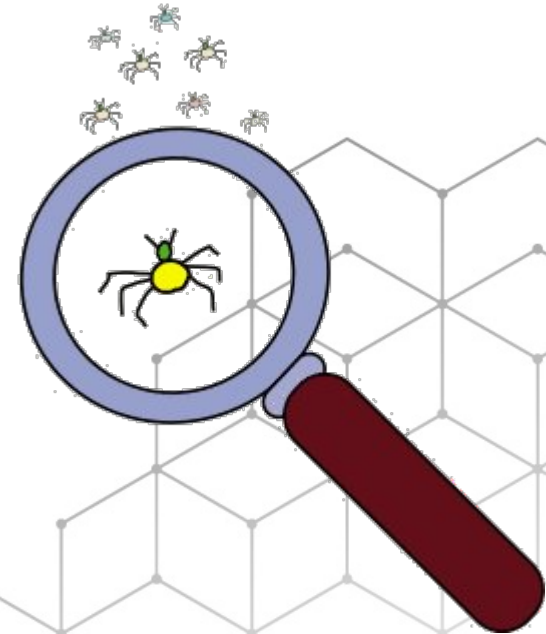


Issues

HDFS is **not optimized for small files**.

Use a more efficient object storage system like Ceph or Delta Lake.

Support data versioning and ACID transactions.

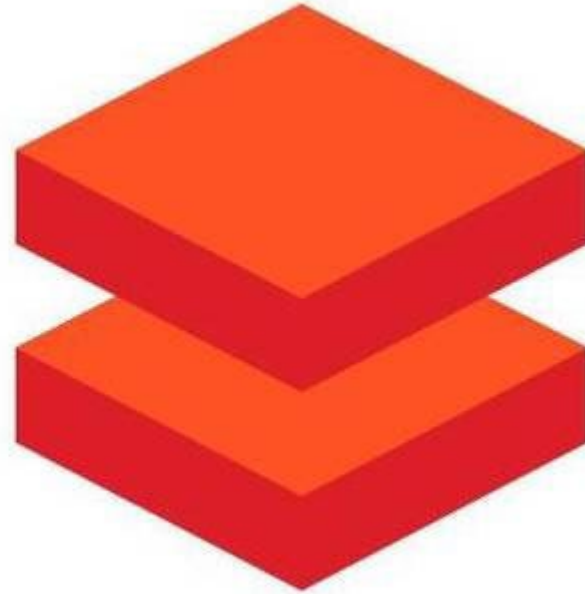


NOVA

IMS

Information
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What do we End Up?



Next Lecture

- Big Data Storage
 - Distributed File System:
 - HDFS
 - BDFS



Questions?

The End

